

Øvelse 1

Exercise 1.4 Consider the problem

$$\begin{array}{ll}\text{minimize} & 2x_1 + 3|x_2 - 10| \\ \text{subject to} & |x_1 + 2| + |x_2| \leq 5,\end{array}$$

and reformulate it as a linear programming problem.

Vi sætter:

$$z_1 = |x_2 - 10| \text{ hvor det gælder at } x_2 - 10 \leq z_1 \text{ og } -x_2 + 10 \leq z_1$$

$$z_2 = |x_1 + 2| \text{ hvor det gælder at } x_1 + 2 \leq z_2 \text{ og } -x_1 - 2 \leq z_2$$

$$z_3 = |x_2| \text{ hvor det gælder at } x_2 \leq z_3 \text{ og } -x_2 \leq z_3$$

Så fås det lineære problem:

$$\min 2x_1 + 3z_1$$

s.t

$$x_2 - 10 \leq z_1$$

$$-x_2 + 10 \leq z_1$$

$$x_1 + 2 \leq z_2$$

$$-x_1 - 2 \leq z_2$$

$$x_2 \leq z_3$$

$$-x_2 \leq z_3$$

$$z_2 + z_3 \leq 5$$

Øvelse 2

Exercise 1.10 (Production and inventory planning) A company must deliver d_i units of its product at the end of the i th month. Material produced during a month can be delivered either at the end of the same month or can be stored as inventory and delivered at the end of a subsequent month; however, there is a storage cost of c_1 dollars per month for each unit of product held in inventory. The year begins with zero inventory. If the company produces x_i units in month i and x_{i+1} units in month $i + 1$, it incurs a cost of $c_2|x_{i+1} - x_i|$ dollars, reflecting the cost of switching to a new production level. Formulate a linear programming problem whose objective is to minimize the total cost of the production and inventory schedule over a period of twelve months. Assume that inventory left at the end of the year has no value and does not incur any storage costs.

Vi lader l_i være den mængde vi har tilbage på lageret i måned i , som må være det vi producerer i måned $i + 1$ minus det vi havde tilbage i måned i minus det vi skal levere. Dvs. at $l_i = x_{i+1} + l_{i-1} - d_i$.

Vi får givet at $l_0 = 0$ og vi har også at $x_i + l_{i-1} \geq d_i$

Altså har vi at:

$$\min c_1 \sum_{i=1}^{11} l_i + c_2 \sum_{i=1}^{11} |x_{i+1} - x_i|$$

s.t

$$l_0 = 0$$

$$l_i = x_i + l_{i-1} - d_i$$

$$x_i + l_{i-1} \geq d_i$$

$$x_i, d_i, l_i \geq 0$$

Vi sætter $z_i = |x_{i+1} - x_i|$ og får følgende lineær programming problem:

$$\min c_1 \sum_{i=1}^{11} l_i + c_2 \sum_{i=1}^{11} z_i$$

s.t

$$l_0 = 0$$

$$l_i = x_i + l_{i-1} - d_i$$

$$x_i + l_{i-1} \geq d_i$$

$$z_i \geq x_{i+1} - x_i$$

$$z_i \geq -x_{i+1} + x_i$$

$$x_i, d_i, l_i \geq 0$$

for $i = 1, \dots, 12$

Øvelse 3

Exercise 1.12 (Chebychev center) Consider a set P described by linear inequality constraints, that is, $P = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_i' \mathbf{x} \leq b_i, i = 1, \dots, m\}$. A ball with center $\mathbf{y}$ and radius r is defined as the set of all points within (Euclidean) distance r from $\mathbf{y}$. We are interested in finding a ball with the largest possible radius, which is entirely contained within the set P . (The center of such a ball is called the *Chebychev center* of P .) Provide a linear programming formulation of this problem.

Vi har her en et vilkårlig polyeder givet ved $P = \{x \in \mathbb{R}^n : a_i^T x \leq b_i, i = 1, \dots, m\}$ og vi ønsker at finde kuglen med den størst mulige radius, således at kuglen er fuldstændigt indeholdt i P .

Denne kugle skal være lukket, ellers findes ikke den største radius.

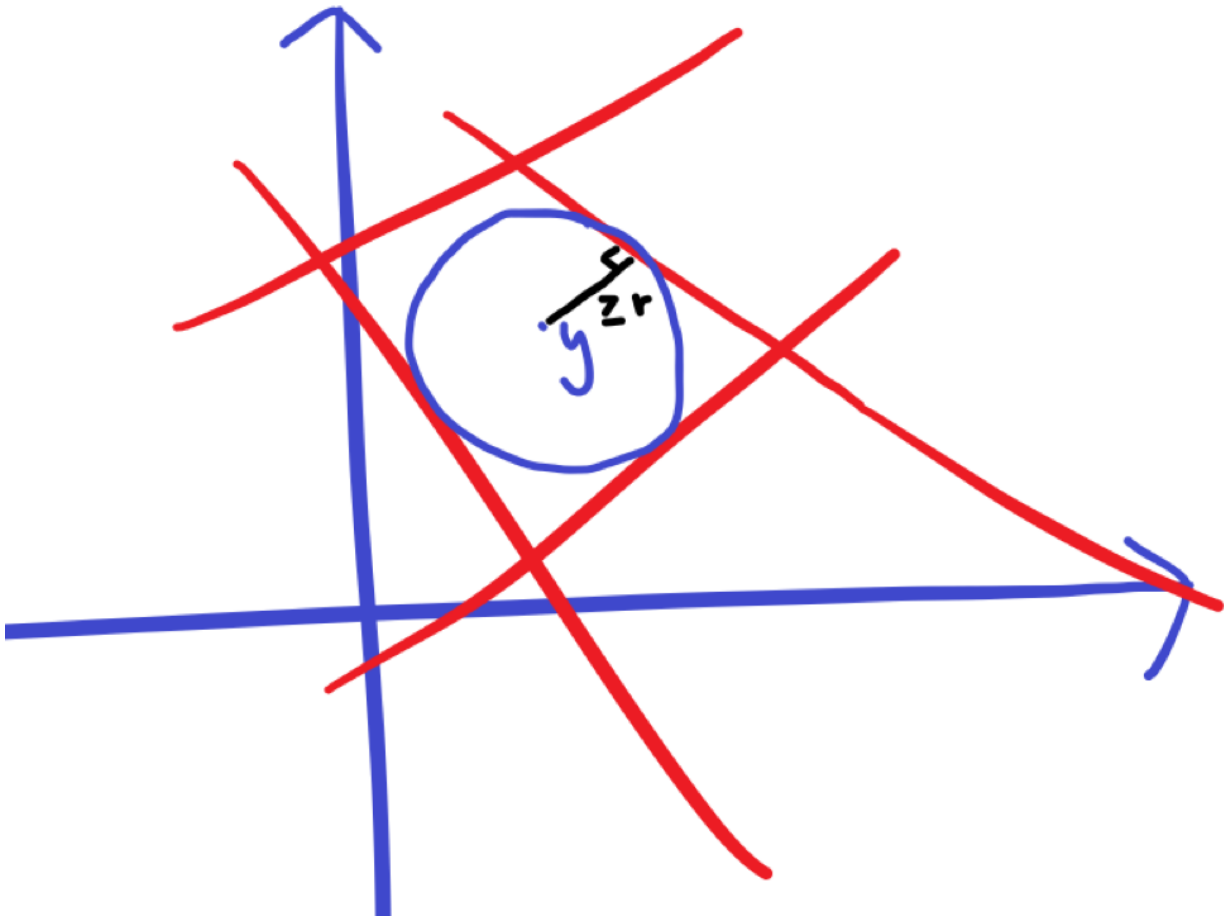
Dvs at kuglen defineres som:

$$B_r(y) = \{x \in \mathbb{R}^n : \|x - y\| \leq r\}$$

Vi ønsker at maksimere r , således at $B_r(y) \subseteq P$, så vi skal finde det bedste punkt i P , som bliver centrum for kuglen, således at vi kan gøre r så stor som mulig og stadig gøre at kuglen forbliver i P .

Alle ligningerne $a_i^T x = b_i$ definerer hyperplaner, og hvis $B_r(y)$ skal være helt indeholdt i P , så skal

afstanden fra centrum y til hver hyperplan være mindst r (og y skal selvfølgelig ligge i P , ellers kunne vi bare ligge y virkeligt langt udenfor P (her findes faktisk ikke en løsning) og stadig opfylde at afstandene er mindst r).



Det er egentligt den MINDSTE afstand fra y til hyperplanerne, som skal være mindst r , og denne mindste afstand er den vinkelrette afstand.

Den vinkelrette afstand mellem y og linjen $a_i^T x = b_i$ er givet ved

$$\frac{|a_i^T y - b_i|}{\|a_i\|} = \frac{b_i - a_i^T y}{\|a_i\|} \quad (\text{da } a_i^T y \leq b_i, \text{ siden } y \text{ også skal ligge i } P)$$

og denne skal være mindst r , dvs

$$\frac{b_i - a_i^T y}{\|a_i\|} \geq r \Leftrightarrow b_i - a_i^T y \geq r \|a_i\| \quad \text{for } i = 1, \dots, m.$$

Altså kan vi formulere følgende problem:

max r

s.t.

$$a_i^T y \leq b_i \quad \text{for } i = 1, \dots, m \quad (\text{sørger for at } y \in P)$$

$$b_i - a_i^T y \geq r \|a_i\| \quad \text{for } i = 1, \dots, m \quad (\text{sørger for at afstanden mellem } y \text{ og hyperplanerne er mindst } r)$$

Øvelse 4

Exercise 1.14 A company produces and sells two different products. The demand for each product is unlimited, but the company is constrained by cash availability and machine capacity.

Each unit of the first and second product requires 3 and 4 machine hours, respectively. There are 20,000 machine hours available in the current production period. The production costs are \$3 and \$2 per unit of the first and second product, respectively. The selling prices of the first and second product are \$6 and \$5.40 per unit, respectively. The available cash is \$4,000; furthermore, 45%

of the sales revenues from the first product and 30% of the sales revenues from the second product will be made available to finance operations during the current period.

- Formulate a linear programming problem that aims at maximizing net income subject to the cash availability and machine capacity limitations.
- Solve the problem graphically to obtain an optimal solution.
- Suppose that the company could increase its available machine hours by 2,000, after spending \$400 for certain repairs. Should the investment be made?

a)

$$\max (6 - 3) x_1 + (5.4 - 2) x_2 = \max 3 x_1 + 3.4 x_2$$

s.t

$$3 x_1 + 4 x_2 \leq 20000$$

$$3 x_1 + 2 x_2 \leq 4000 + (6 \cdot 0.45) x_1 + (5.4 \cdot 0.3) x_2$$

$$\Leftrightarrow 0.3 x_1 + 0.38 x_2 \leq 4000$$

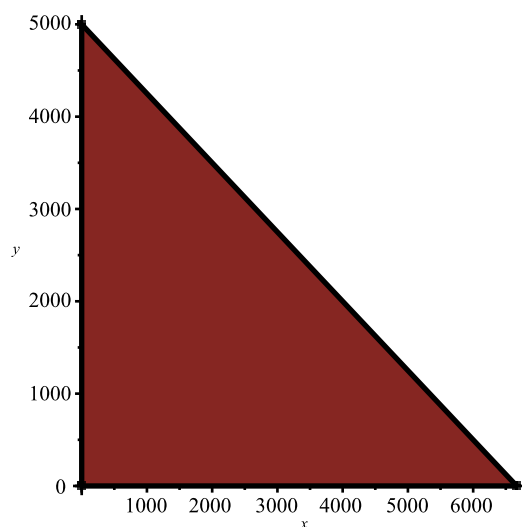
$$x_1, x_2 \geq 0$$

b) Grafisk løsning:

with(PolyhedralSets) :

$$P := \text{PolyhedralSet} \left(\left[3 \cdot x + 4 \cdot y \leq 20000, \frac{3}{10} \cdot x + \frac{38}{100} \cdot y \leq 4000, x \geq 0, y \geq 0 \right] \right) :$$

Plot(P)



Optimal løsning er $(x_1^*, x_2^*) = \left(\frac{20000}{3}, 0 \right)$ som giver profit 20000.

c) Vi får nu følgende maksimeringsproblem:

$$\max 3x_1 + 3.4x_2$$

s.t

$$3x_1 + 4x_2 \leq 22000$$

$$0.3x_1 + 0.38x_2 \leq 4000 - 400 = 3600$$

$$x_1, x_2 \geq 0$$

som giver $(x_1^*, x_2^*) = \left(\frac{22000}{3}, 0\right)$ og net income på 22000, så det er en god investering, da profitten stiger med 2000 og outlay kun er 400.

Øvelse 5

Exercise 1.15 A company produces two kinds of products. A product of the first type requires $1/4$ hours of assembly labor, $1/8$ hours of testing, and \$1.2 worth of raw materials. A product of the second type requires $1/3$ hours of assembly, $1/3$ hours of testing, and \$0.9 worth of raw materials. Given the current personnel of the company, there can be at most 90 hours of assembly labor and 80 hours of testing, each day. Products of the first and second type have a market value of \$9 and \$8, respectively.

- (a) Formulate a linear programming problem that can be used to maximize the daily profit of the company.
- (b) Consider the following two modifications to the original problem:
- (i) Suppose that up to 50 hours of overtime assembly labor can be scheduled, at a cost of \$7 per hour.
 - (ii) Suppose that the raw material supplier provides a 10% discount if the daily bill is above \$300.

Which of the above two elements can be easily incorporated into the linear programming formulation and how? If one or both are not easy to incorporate, indicate how you might nevertheless solve the problem.

a) Problemet bliver:

$$\max \cdot (9 - 1.2)x_1 + (8 - 0.9)x_2$$

s.t

$$\frac{1}{4}x_1 + \frac{1}{3}x_2 \leq 90 \text{ (labor)}$$

$$\frac{1}{8}x_1 + \frac{1}{3}x_2 \leq 80 \text{ (testing)}$$

$$x_1, x_2 \geq 0$$

b)

i) Lad T betegne antallet af overtimer og tilføj den i labor begrænsningen samt som omkostning i objektfunktionen.

Så fås maksimeringsproblemet til:

$$\max \cdot (9 - 1.2)x_1 + (8 - 0.9)x_2 - 7T$$

s.t

$$\frac{1}{4}x_1 + \frac{1}{3}x_2 \leq 90 + T \text{ (labor)}$$

$$\frac{1}{8}x_1 + \frac{1}{3}x_2 \leq 80 \text{ (testing)}$$

$$T \leq 50$$

$$x_1, x_2, T \geq 0$$

Dette kan som vist nemt implementeres i det lineære program, og problemet kan let løses ved sædvanlig vis.

ii) Her fås en 10% rabat, hvis der købes mere end 300 på materialer.

Dvs, hvis $1.2 x_1 + 0.9 x_2 \geq 300$, så skal prisen ganges med 0.9.

Dette kan ikke nemt implementeres.

Her kan man starte med at løse det originale problem og finde optimum og derefter sammenligne det med problemet hvor vi ganger prisen med 0.9 og samt tilføjer begrænsningen $1.2 x_1 + 0.9 x_2 \geq 300$, for at tvinge at rabatten er brugt.

Så kan man sammenligne de to optimale løsninger og vælge den bedste.

Altså må dette problem løses vha. 2 lineære programmer.